

MIT integration bee - 2020

Notación y Consideraciones Generales

A lo largo del documento vamos a utilizar la siguiente notación, en algunos problemas más complejos utilizaremos la notación usual:

$$\{ f(x) \}_a^b := \int_a^b f(x) dx, \quad [f(x)] := \frac{d}{dx} f(x)$$

Recursos obtenidos de: <https://math.mit.edu/~yyao1/integrationbee.html>

Herramientas que te pueden ser útiles:

- Calculadora gráfica: <https://www.desmos.com/calculator?lang=es>
- Calculadora de integrales: <https://mathdf.com/es/>
- Calculadora de integrales: <https://www.wolframalpha.com/>
- Motor de búsqueda para fórmulas en latex <https://approach0.xyz/search/>
- Te ayuda con ideas: <https://chat.deepseek.com>

Ejercicio 1

$$\int \frac{\log(2x)}{x \log x} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{\ln(2) + \ln(x)}{x \ln x} \right\} \\ &= \left\{ \frac{\ln(2) + t}{t} \right\} \quad (1) \\ &= \ln(2) \ln |t| + t + C \\ &= \ln(2) \ln |\ln(x)| + \ln(x) + C \end{aligned}$$

$$\bullet (1): \begin{cases} t = \ln(x) \\ dt = \frac{1}{x} dx \end{cases}$$

Ejercicio 2

$$\int_0^\infty \frac{1}{e^x + 1} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ \frac{e^u}{e^u + 1} \right\}_{-\infty}^0 \quad (1) \\ &= (\ln |e^u + 1|)_{-\infty}^0 \\ &= \ln(2) \end{aligned}$$

$$\bullet (1): \begin{cases} x = -u \\ dx = -du \end{cases}$$

Ejercicio 3

$$\int_e^{e^e} \frac{\log x \cdot \log(\log x)}{x} dx$$

— — — Demostración — — —

$$I = \{ t \ln(t) \}_1^e \quad (1)$$

$$= \frac{1}{2} (t^2 \ln(t))_1^e - \frac{1}{2} \{ t \}_1^e \quad (2)$$

$$= \frac{e^2}{2} - \frac{1}{2} \left(\frac{t^2}{2} \right)_1^e$$

$$= \frac{e^2}{4} + \frac{1}{4}$$

$$\bullet (1): \begin{cases} t = \ln(x) \\ dt = \frac{1}{x} dx \end{cases}$$

$$\bullet (2): \left| \begin{array}{c|c} D & I \\ \ln(t) & t \\ \frac{1}{t} & \frac{t^2}{2} \end{array} \right|$$

Ejercicio 4

$$\int_0^1 \log \left(\frac{1+x}{1-x} \right) dx$$

— — — Demostración — — —

$$I_1 = x \ln \left(\frac{1+x}{1-x} \right) - 2 \left\{ \frac{x}{1-x^2} \right\} \quad (1)$$

$$= x \ln \left(\frac{1+x}{1-x} \right) + \ln |1-x^2| + C \quad (2)$$

$$\begin{aligned} I &= (x \ln \left(\frac{1+x}{1-x} \right) + \ln |1-x^2|)_0^1 \\ &= (x \ln ((1+x)^2))_0^1 \\ &= \ln(4) \end{aligned}$$

$$\bullet (1): \left| \begin{array}{c|c} D & I \\ \ln \left(\frac{1+x}{1-x} \right) & dx \\ \frac{2}{1-x^2} & x \end{array} \right|$$

$$\bullet (2): [\ln(f(x))] = \frac{f'(x)}{f(x)}$$

Nota:

1. Al aplicar integración por partes debemos de tener cuidado con las integrales impropias:

$$I = \left(x \ln \left(\frac{1+x}{1-x} \right) \right)_0^1 - 2 \left\{ \frac{x}{1-x^2} \right\}_0^1$$

Ejercicio 5

$$\int \frac{1}{x^2 + (x-1)^2} dx$$

— — — Demostración — — —

$$\begin{aligned}
 I &= - \left\{ \frac{1}{1 + (1 - t)^2} \right\} \quad (1) \\
 &= \arctan(1 - t) + C \\
 &= \arctan \left(1 - \frac{1}{x} \right) + C
 \end{aligned}$$

$$\bullet (1): \begin{cases} x &= \frac{1}{t} \\ dx &= -\frac{1}{t^2} dt \end{cases}$$

Ejercicio 6

$$\int \sqrt{x \sqrt{x \cdots x}} dx$$

--- Demostración ---

$$\begin{aligned}
 I &= \left\{ x^{1/2} x^{1/4} x^{1/8} \cdots \right\} \\
 &= \left\{ x^{\sum_{n \geq 1} (1/2)^n} \right\} \\
 &= \left\{ x^{\sum_{n \geq 0} (1/2)^{n-1}} \right\} \\
 &= \left\{ x^{2-1} \right\} \quad (1) \\
 &= \frac{x^2}{2} + C
 \end{aligned}$$

$$\bullet (1): \text{Serie geométrica } \sum_{n \geq 0} r^n = \frac{1}{1-r} \text{ con } |r| < 1$$

Ejercicio 7

$$\int \sin^4 x \cos^4 x (\cos x + \sin x)(\cos x - \sin x) dx$$

--- Demostración ---

$$\begin{aligned}
 I &= \left\{ \sin^4 x \cos^4 x (\cos^2 x - \sin^2 x) \right\} \\
 &= \left\{ \sin^4 x \cos^4 x \cos(2x) \right\} \quad (1) \\
 &= \frac{1}{2^4} \left\{ (2 \sin x \cos x)^4 \cos(2x) \right\} \\
 &= \frac{1}{2^4} \left\{ \sin^4(2x) \cos(2x) \right\} \quad (2) \\
 &= \frac{1}{2^5} \frac{\sin^5(2x)}{5} + C \\
 &= \frac{\sin^5(x) \cos^5(x)}{5} + C
 \end{aligned}$$

$$\bullet (1): \cos(2x) = \cos^2(x) - \sin^2(x)$$

$$\bullet (2): \sin(2x) = 2 \sin(x) \cos(x)$$

Ejercicio 8

$$\int \log(x^2 + 1) dx$$

--- Demostración ---

$$\begin{aligned}
 I &= x \ln(1+x^2) - 2 \left\{ \frac{x^2}{1+x^2} \right\} \\
 &= x \ln(1+x^2) - 2 \left\{ 1 - \frac{1}{1+x^2} \right\} \\
 &= x \ln(1+x^2) - 2(x - \arctan(x)) + C
 \end{aligned} \tag{1}$$

$$\bullet (1): \left| \ln(1+x^2) \right| \left| \frac{D}{\frac{2x}{1+x^2}} \right| \left| \frac{I}{x} \right|$$

Ejercicio 9

$$\int_0^{2\pi} \cos^{2020}(x) dx$$

--- Demostración ---

$$I = \frac{2\pi}{2^{2020}} \binom{2020}{1010} \tag{1}$$

- (1): Usando variable compleja tenemos el siguiente resultado:

[https://math.stackexchange.com/questions/1366304/integration-of-int-02-pi-cos2ntdt?](https://math.stackexchange.com/questions/1366304/integration-of-int-02-pi-cos2ntdt?noredirect=1)
[noredirect=1](#)

$$\int_0^{2\pi} \cos^{2n}(x) dx = \frac{2\pi}{2^{2n}} \binom{2n}{n}$$

Ejercicio 10

$$\int \frac{2x+1}{2x^2+2x+1} dx$$

--- Demostración ---

$$\begin{aligned}
 I &= \frac{1}{2} \left\{ \frac{4x+2}{2x^2+2x+1} \right\} \\
 &= \frac{1}{2} \ln|2x^2+2x+1| + C \tag{1}
 \end{aligned}$$

$$\bullet (1): [\ln(f(x))] = \frac{f'(x)}{f(x)}$$

Ejercicio 11

$$\int_{1/\sqrt{2}}^1 \frac{\arcsin(x)}{x^3} dx$$

--- Demostración ---

$$\begin{aligned}
I &= -\frac{1}{2} \left(\frac{\arcsin(x)}{x^2} \right)^1 + \frac{1}{2} \left\{ \frac{1}{x^2 \sqrt{1-x^2}} \right\}_{1/\sqrt{2}}^1 \quad (1) \\
&= \frac{1}{2} \left\{ \frac{1}{x^2 \sqrt{1-x^2}} \right\}_{1/\sqrt{2}}^1 \\
&= \frac{1}{2} \left\{ \csc^2(\theta) \right\}_{\pi/4}^{\pi/2} \quad (2) \\
&= -\frac{1}{2} \left(\cot(\theta) \right)_{\pi/4}^{\pi/2} \\
&= \frac{1}{2}
\end{aligned}$$

- (1): $\left| \begin{array}{c} D \\ \arcsin(x) \\ \frac{1}{\sqrt{1-x^2}} \end{array} \right| \left| \begin{array}{c} I \\ \frac{1}{x^3} \\ -\frac{1}{2x^2} \end{array} \right|$
- (2): $\left\{ \begin{array}{l} x = \sin(\theta) \\ dx = \cos(\theta) d\theta \end{array} \right.$

Ejercicio 12

$$\int_0^{\pi/2} \sin(2x) \cos(\cos(x)) dx$$

— — — Demostración — — —

$$I = 2 \left\{ \sin(x) \cos(x) \cos(\cos(x)) \right\}_0^{\pi/2} \quad (1)$$

$$= 2 \left\{ u \cos(u) \right\}_0^1 \quad (2)$$

$$= 2 \left(u \sin(u) + \cos(u) \right)_0^1 \quad (3)$$

$$= 2 \left(\sin(1) + \cos(1) - 1 \right)$$

- (1): $\sin(2x) = 2 \sin(x) \cos(x)$

- (2): $\left\{ \begin{array}{l} u = \cos(x) \\ du = -\sin(x) dx \end{array} \right.$

- (3): $\left| \begin{array}{c} D \\ u \\ 1 \\ 0 \end{array} \right| \left| \begin{array}{c} I \\ \cos(u) \\ \sin(u) \\ -\cos(u) \end{array} \right|$

Ejercicio 13

$$\int_0^{2\pi} \sin(\sin(x) - x) dx$$

— — — Demostración — — —

$$I = \{ \sin(\sin(x) - x) \}_0^\pi + \{ \sin(\sin(x) - x) \}_\pi^{2\pi}$$

$$\begin{aligned} I_1 &= \{ \sin(\sin(x + \pi) - (x + \pi)) \}_0^\pi \\ &= \{ \sin(-(\sin(x) + x + \pi)) \}_0^\pi \end{aligned} \quad (1)$$

$$= \{ \sin(\sin(x) + x + \pi) \}_0^\pi \quad (2)$$

$$= \{ \sin(\sin(x) + x) \}_0^\pi \quad (1)$$

$$= \{ \sin(\sin(x)) \cos(x) + \cos(\sin(x)) \sin(x) \}_0^\pi \quad (3)$$

$$\begin{aligned} I_2 &= \{ \sin(\sin(x) - x) \}_0^\pi \\ &= \{ \sin(\sin(x)) \cos(x) - \cos(\sin(x)) \sin(x) \}_0^\pi \end{aligned}$$

$$\begin{aligned} I &= 2 \{ \sin(\sin(x)) \cos(x) \}_0^\pi \\ &= -2 \{ \cos(\sin(x)) \}_0^\pi \\ &= 0 \end{aligned} \quad (4)$$

- (1): $\sin(x + \pi) = -\sin(x)$
- (2): Función impar
- (3): $\sin(a \pm b) = \sin(a) \cos(b) \pm \cos(a) \sin(b)$
- (4): $[f(g(x))] = f'(g(x))g'(x)$

Ejercicio 14

$$\int \frac{1}{x-1} + \frac{\sum_{k=0}^{2018} (k+1)x^k}{\sum_{k=0}^{2019} x^k} dx$$

--- Demostración ---

$$\begin{aligned} I &= \left\{ \frac{1}{x-1} + \frac{\sum_{k=0}^{\alpha-1} (k+1)x^k}{\sum_{k=0}^{\alpha} x^k} \right\} \\ &= \ln|x-1| + \left\{ \frac{\sum_{k=0}^{\alpha-1} (k+1)x^k}{\sum_{k=0}^{\alpha} x^k} \right\} \\ &= \ln|x-1| + \ln \left| \sum_{k=0}^{\alpha} x^k \right| + C \quad (1) \\ &= \ln|x-1| + \ln \left| \frac{x^{\alpha+1} - 1}{x-1} \right| + C \quad (2) \\ &= \ln|x^{\alpha+1} - 1| + C \quad (2) \end{aligned}$$

- (1): $[\sum_{k=0}^{\alpha} x^k] = [1 + \sum_{k=1}^{\alpha} x^k] = \sum_{k=1}^{\alpha} kx^{k-1} = \sum_{k=0}^{\alpha-1} (k+1)x^k$
- (2): $a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + b^{n-2}a + b^{n-1})$

Ejercicio 15

$$\int_0^{\frac{\pi}{2}} \frac{1}{\tan^{\sqrt{2020}}(x) + 1} dx$$

--- Demostración ---

$$\begin{aligned}
I &= \left\{ \frac{1}{\tan^\alpha(x) + 1} \right\}_0^{\pi/2} \\
&= \left\{ \frac{1}{\cot^\alpha(\theta) + 1} \right\}_0^{\pi/2} \quad (1), (2)
\end{aligned}$$

$$\begin{aligned}
2I &= \left\{ \frac{1}{\tan^\alpha(x) + 1} + \frac{1}{\cot^\alpha(x) + 1} \right\}_0^{\pi/2} \\
&= \left\{ \frac{\cos^\alpha(x)}{\sin^\alpha(x) + \cos^\alpha(x)} + \frac{\sin^\alpha(x)}{\sin^\alpha(x) + \cos^\alpha(x)} \right\}_0^{\pi/2} \\
&= \left\{ 1 \right\}_0^{\pi/2} \\
&= \pi/2
\end{aligned}$$

$$I = \pi/4$$

- (1): $\begin{cases} x &= \pi/2 - \theta \\ dx &= -d\theta \end{cases}$
- (2): $\tan(\pi/2 - x) = \cot(x)$

Ejercicio 16

$$\int x(1-x)^{2020} dx$$

--- Demostración ---

$$\begin{aligned}
I &= - \left\{ (1-u)u^\alpha \right\} \quad (1) \\
&= - \frac{u^{\alpha+1}}{\alpha+1} + \frac{u^{\alpha+2}}{\alpha+2} + C
\end{aligned}$$

- (1): $\begin{cases} u &= 1-x \\ du &= -dx \end{cases}$

Ejercicio 17

$$\int \frac{\sec^4(x) \tan(x)}{\sec^4(x) + 4} dx$$

--- Demostración ---

$$\begin{aligned}
I &= \frac{1}{4} \left\{ \frac{4 \sec^3(x) \sec(x) \tan(x)}{\sec^4(x) + 4} \right\} \\
&= \frac{1}{4} \ln |\sec^4(x) + 4| + C \quad (1), (2)
\end{aligned}$$

- (1) $[\sec^4(x)] = 4 \sec^4(x) \tan(x)$
- (2) $[\ln(f(x))] = \frac{f'(x)}{f(x)}$

Ejercicio 18

$$\int x^{2x} (2 \log(x) + 2) dx$$

--- Demostración ---

$$I = x^{2x} + C \quad (1)$$

- (1): $[x^{2x}] = [e^{2x \ln(x)}] = e^{2x \ln(x)}(2 \ln(x) + 2) = x^{2x}(2 \log(x) + 2)$

Ejercicio 19

$$\int_0^1 \sqrt{1-x^2} dx$$

— — — Demostración — — —

Sea $y = \sqrt{1-x^2}$, entonces $y^2 + x^2 = 1$ que representa la ecuación de una circunferencia de radio $r = 1$ y centro en $(0, 0)$. Así pues, la integral está calculando una cuarta parte del área de esta circunferencia.

$$\begin{aligned} I &= \frac{\pi r^2}{4} \\ &= \frac{\pi}{4} \end{aligned}$$

Ejercicio 20

$$\int_0^\infty x^5 e^{-x^4} dx$$

— — — Demostración — — —

$$\begin{aligned} I &= \left\{ x^3 x^2 e^{-x^4} \right\}_0^\infty \\ &= \frac{1}{4} \left\{ u^{1/2} e^{-u} \right\}_0^\infty \quad (1) \\ &= \frac{1}{4} \Gamma(3/2) \\ &= \frac{1}{8} \Gamma(1/2) \quad (3) \\ &= \frac{\sqrt{\pi}}{8} \quad (4) \end{aligned}$$

- (1): $\begin{cases} u &= x^4 \\ du &= 4x^3 dx \end{cases}$

- (2): Funcion Gamma $\Gamma(n) = \{t^{n-1} e^{-t}\}_0^\infty$

- (3): $\Gamma(x+1) = x\Gamma(x)$

- (4): $\Gamma(1/2) = \sqrt{\pi}$ que se puede deducir de la integral Gaussiana $\{e^{-x^2}\}_{-\infty}^\infty = \sqrt{\pi}$